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Editor
The Journal of Chemical Physics

Dear Editor,

Please find enclosed the manuscript "Algorithmic Construction and Exploitation of Topology-Constrained Refinement Models for Semiexperimental Equilibrium Structures," submitted for consideration as an Article in *The Journal of Chemical Physics*.

The manuscript introduces MORPHEUS, a framework that turns the formulation of semiexperimental refinement models into a reproducible topology-, symmetry-, and rank-audited procedure. Rather than assuming that the internal-coordinate model has already been selected by expert intervention, MORPHEUS uses SMITH-generated SONIC coordinate contracts, applies symmetry and homogeneous rank reduction, projects constraints and predicates consistently, and propagates uncertainties back to chemically interpretable structural parameters. When an internal-coordinate correction must be realized, MORPHEUS delegates that operation to the LINK geometry service rather than owning a second Wilson-B inversion algorithm.

The work is, in my view, well aligned with the scope of *The Journal of Chemical Physics* because it addresses a general problem at the interface of rotational spectroscopy, molecular structure determination, and quantum chemistry: how to combine high-resolution experimental data, computed vibrational and electronic corrections, topology, symmetry, and external reference information in a controlled statistical model. The validation spans acyclic, cyclic, planar, fused, bridged, and flexible molecules. A large-molecule testosterone case study further shows that accurate quantum chemical equilibrium structures, accelerated vibrational corrections, and algorithmically generated topology-constrained refinement models can be chained for systems containing about 50 atoms, where conventional isotope-rich semiexperimental analyses are not currently practical.

The manuscript also emphasizes reproducibility. The reported models are not defined by molecule-specific Z-matrix choices, but by explicit primitive relations, parameter classes, predicates, and audit trails. The submitted Supporting Information documents the benchmark inputs, diagnostics, and the large-molecule class/predicate variants used in the paper.

The coordinate-generation layer used here is the subject of a companion SMITH/SONIC methodological preprint. If useful for evaluation, that manuscript can be supplied as confidential supplementary material for the referees.

This manuscript is original, has not been published previously, and is not under consideration elsewhere. I confirm that there are no conflicts of interest to declare.

Sincerely,

Vincenzo Barone